

SYLLABUS
FINANCIAL TIME SERIES
46-929

INSTRUCTOR: John Lehoczky
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CLASS PERIOD: Tuesday and Thursday, 1:30-3:00pm (EDT)
TA SESSIONS: Friday 11:00-12:00 (EDT)
COURSE ASSISTANT: Wanshan Li, wanshanl@andrew.cmu.edu
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CANVAS: S21-46929
FINAL EXAM: Monday, May 10, 2021, 10:00am (EDT)
OFFICE HOURS: By Arrangement Using Zoom

Course Description

This course is designed for students in the Carnegie Mellon MSCF program who have completed the Financial Data Science sequence. There will be modules on

- The stylized facts for equities and other financial markets
- Stationarity
- Basic time series models including autoregressive (AR), moving average (MA), ARMA and ARIMA. Transformations to stationarity.
- Prediction
- Multivariate Time Series (Vector autoregressive processes, Dickey Fuller test for stationarity and cointegration)
- ARCH and GARCH modeling with applications to risk management
- State space models and Kalman filtering

Course Learning Objectives

At the completion of this course, all students should

1. Have a working knowledge the role time series models plays in computational finance.
2. Have a good command of elementary time series concepts including stationarity and the basic time series models.
3. Have a good understanding of more complex models such as GARCH and state space modeling.
4. Gain expertise in writing Python programs to implement time series techniques.

5. Be well prepared for summer internships and future employment that may involve time series analysis.

Course Prerequisites

It is assumed that students have taken and passed the Carnegie Mellon MSCF courses Probability Prep and Data Science 1 and 2. In addition students must have competence in python equivalent to that covered in FC 1 and 2.

Course Materials

There is no textbook required for the course; however, there is a large set of course notes that will cover the course material, in some cases in greater depth. In addition, there are a number of excellent reference texts listed below.

- Box, G.E.P., Jenkins, G., Reinsel, G. and G.M. Ljung, “*Time Series Analysis: Forecasting and Control, 5th edition*, John Wiley, 2015.
- Brockwell, P.J. and R. Davis, *Introduction to Time Series and Forecasting, 3rd Edition*, Springer, 2016.
- Chan, N.H., *Time Series: Applications to Finance with R and S-Plus*, Wiley, 2010.
- Hamilton, James, *Time Series Analysis*, Princeton University Press, 1994.
- Lai, T.L. and Xing, H., *Statistical Models and Methods for Financial Markets*, Springer, 2008.
- McKinney, W., *Python for Data Analysis: Data Wrangling with Pandas, Numpy and IPython, 2nd Edition*, O’Reilly, 2018.
- Ruppert, D. and D. Matteson, *Statistics and Data Analysis for Financial Engineering, 2nd Edition*, Springer, 2015.
- Sargent, T.J. and J. Stachurski, *Quantitative Economics with Python*, 2020. On-line <http://python.quantecon.org> .
- Tsay, R.S., *An Introduction to the Analysis of Financial Data with R*, Wiley, 2013.
- Tsay, R.S., *Analysis of Financial Time Series, 3rd Edition*, Wiley, 2019.
- Tsay, R.S. *Multivariate Time Series Analysis with R and Financial Applications*, Wiley, 2014.
- Velu, R., Hardy, M. and D. Nehren, *Algorithmic Trading and Quantitative Strategies*, CRC Press, 2020.

Required Work and Grading

The class coursework will consist of roughly 5 or 6 weekly homework assignments and a final exam. Homeworks should be uploaded as one pdf file or Jupyter notebook, to Canvas.

1. **All homework assignments are due at 1:30pm on the due date.** Indicate clearly on the front of your solutions whether you are a Pittsburgh or a NY student.
2. Late homework is not permitted except in unusual circumstances and by prior arrangement. Slightly late papers (i.e. papers that do not meet the deadline but are submitted before solutions are posted) will be graded, but students will suffer a lateness penalty. If a student cannot finish an assignment by the deadline, that student should submit all completed problems for full credit, then submit the remaining solutions slightly late. Only the problems submitted late will receive a lateness penalty. Once the solutions for an assignment are posted, submissions for that assignment will no longer be accepted.
3. Historically and especially recently there has been a large amount of unwarranted collaboration by MSCF students, sometimes even violating cheating policies. Every year in this simulation class there is a set of students who earn near perfect grades on homework but who perform poorly on the final exam. This year, I will put equal weight on the final exam (50%) and on the homework (50%). Mastery of the homework will ensure a good final exam performance and a good course grade. Lack of effort on the homework will likely result in a poor final exam performance and course grade. More importantly, it will compromise your knowledge of an important computational finance topic, namely time series analysis. Also please note the Policy on Collaboration for this course.

In addition, all MSCF students should recognize and accept their responsibility to help the MSCF faculty to improve the MSCF program and their individual courses. It is imperative that every MSCF student participate conscientiously in the course evaluation process.

Programming Language

This course will be Python-based, as Python is now the dominant language used in the financial services industry and it is very suitable for time series analysis. Furthermore, it will give students experience on using Python to solve financial problems.

Policy on Collaboration

It is important that every student complete this course with a full working knowledge of basic time series analysis. Such a level of mastery can be achieved by every student, but only if each student:

- does the required reading,
- attends the lectures and engages in the material, and
- does all the homework and works essentially independently on those assignments.

It is important that each student make an **individual** effort on each of the homework exercises. Once a student has made a serious individual effort on a problem, he or she should seek help from the instructor, TA or fellow students. **No student should ever seek help, and another student should never offer help, until he or she has made an effort to solve the problem and not succeeded.**

All students must write up their work independently. Computer programs and output must be written and generated independently by each student. All work submitted for credit by any student will be assumed to carry that student's pledge of compliance with this policy on collaboration. Any violation of this policy will result in a grade reduction, possible failure in the course, and the incident will be turned over to the MSCF Steering Committee for possible further penalty.

Intellectual Property

All posted course materials and any recordings of any course sessions are provided solely for educational use by students enrolled in the course and may not be distributed to any other person or posted on the internet without the express written permission of the course instructor.

Accommodating Students with Disabilities

If you have a disability and have an accommodation letter from the Disability Resources office, please contact me to discuss your accommodations and needs as early in the semester as possible. I will work with you to ensure that appropriate accommodations are provided. If you suspect that you may have a disability but are not yet registered with the Office of Disability Resources, I encourage you to contact them at access@andrew.cmu.edu. You may also visit their website, <https://www.cmu.edu/disability-resources/>

Support for Students' Health and Well-being

The combination of challenging academic classes and job searching can become very stressful for MSCF students. Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising (more than your fingers on a keyboard), avoid drugs and alcohol, get enough sleep, and take some time to relax. This will help you achieve your goals and cope with the inevitable stress.

All of us benefit from support during times of stress. There are many helpful resources at CMU that can help. The most important ones are nearby, your academic advisors Jess and Diffy. When help is needed, it is important to ask for help rather than suffering in silence. Also, if you are aware of any fellow students who would benefit from help, please try to facilitate their getting the help they need.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, I strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is available in Pittsburgh (call 412-268-2922) and you can visit their website at <http://www.cmu.edu/counseling/>. Diffy can connect you with an equivalent service in New York.

Policy on Diversity

It is my intent that students from all diverse backgrounds and perspectives be well served by this course, that students learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. It is my intent to be fully respectful of the many dimensions of

diversity including: gender, sexuality, disability, age, socioeconomic status, ethnicity, race, and culture, and this applies to all students as well. Your suggestions are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or student groups. In addition, if any of our class meetings conflict with your religious events, please let me know so that we can make arrangements for you.